

Class 10: Structural Bioinformatics 1

Brian Wong (PID: A18639001)

Table of contents

Introduction to the RCSB Protein Data Bank (PDB)	1
PDB Statistics	2
Visualizing the HIV-1 Protease Structure	3
Bio3D Package for Structural Bioinformatics	4
Predicting Functional Motions of a Single Structure	6

Introduction to the RCSB Protein Data Bank (PDB)

The Protein Data Bank (PDB) is the main repository of biomolecular structures. Let's see what it contains:

```
stats <- read.csv("Data Export Summary.csv")
stats
```

	Molecular.Type	X.ray	EM	NMR	Integrative	Multiple.methods
1	Protein (only)	178,795	21,825	12,773	343	226
2	Protein/Oligosaccharide	10,363	3,564	34	8	11
3	Protein/NA	9,106	6,335	287	24	7
4	Nucleic acid (only)	3,132	221	1,566	3	15
5	Other	175	25	33	4	0
6	Oligosaccharide (only)	11	0	6	0	1
	Neutron Other Total					
1	84 32	214,078				
2	1 0	13,981				
3	0 0	15,759				
4	3 1	4,941				
5	0 0	237				
6	0 4	22				

The comma in these numbers leads to the numbers here being read as characters.

PDB Statistics

```
library(readr)
stats <- read_csv("Data Export Summary.csv")
```

Rows: 6 Columns: 9

-- Column specification -----

Delimiter: ","

chr (1): Molecular Type

dbl (4): Integrative, Multiple methods, Neutron, Other

num (4): X-ray, EM, NMR, Total

i Use `spec()` to retrieve the full column specification for this data.

i Specify the column types or set `show\_col\_types = FALSE` to quiet this message.

```
stats
```

A tibble: 6 x 9

`Molecular Type` <chr>	`X-ray` <dbl>	EM <dbl>	NMR <dbl>	Integrative <dbl>	`Multiple methods` <dbl>	Neutron <dbl>
1 Protein (only)	178795	21825	12773	343	226	84
2 Protein/Oligosacch~	10363	3564	34	8	11	1
3 Protein/NA	9106	6335	287	24	7	0
4 Nucleic acid (only)	3132	221	1566	3	15	3
5 Other	175	25	33	4	0	0
6 Oligosaccharide (o~	11	0	6	0	1	0

i 2 more variables: Other <dbl>, Total <dbl>

Q1: What percentage of structures in the PDB are solved by X-Ray and Electron Microscopy.

80.95% of the structures in the PDB are solved by X-Ray while 12.84% of the structures are solved by EElectron Microscopy.

```
n.xray <- sum(stats$`X-ray`)
n.em <- sum(stats$EM)
n.total <- sum(stats$Total)

n.xray / n.total
```

```
[1] 0.8095077
```

```
n.em / n.total
```

```
[1] 0.1283843
```

Q2: What proportion of structures in the PDB are protein?

97.91% of the structures in the PDB are proteins.

```
sum(stats[1:3,"Total"]) / sum(stats$Total)
```

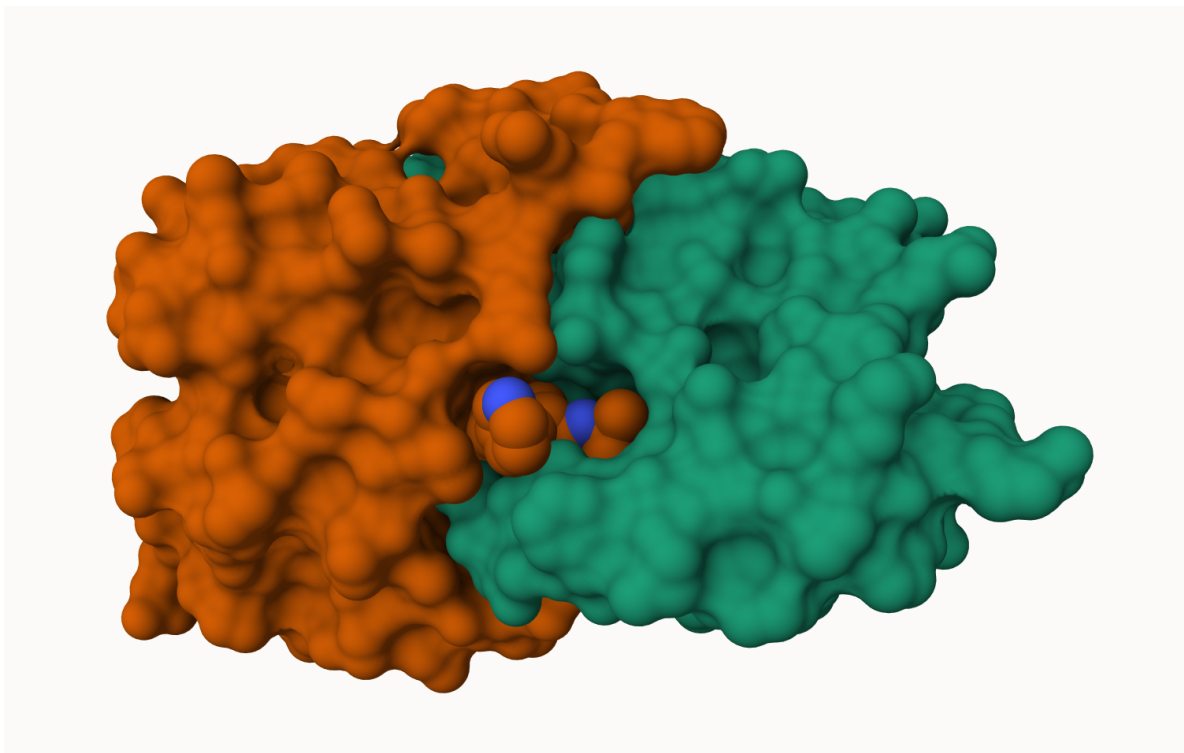
```
[1] 0.979118
```

Q3: Type HIV in the PDB website search box on the home page and determine how many HIV-1 protease structures are in the current PDB?

There are currently 1173 HIV-1 protease structures in the current PDB.

Visualizing the HIV-1 Protease Structure

We can use the Molstar viewer online: <https://molstar.org/viewer/>

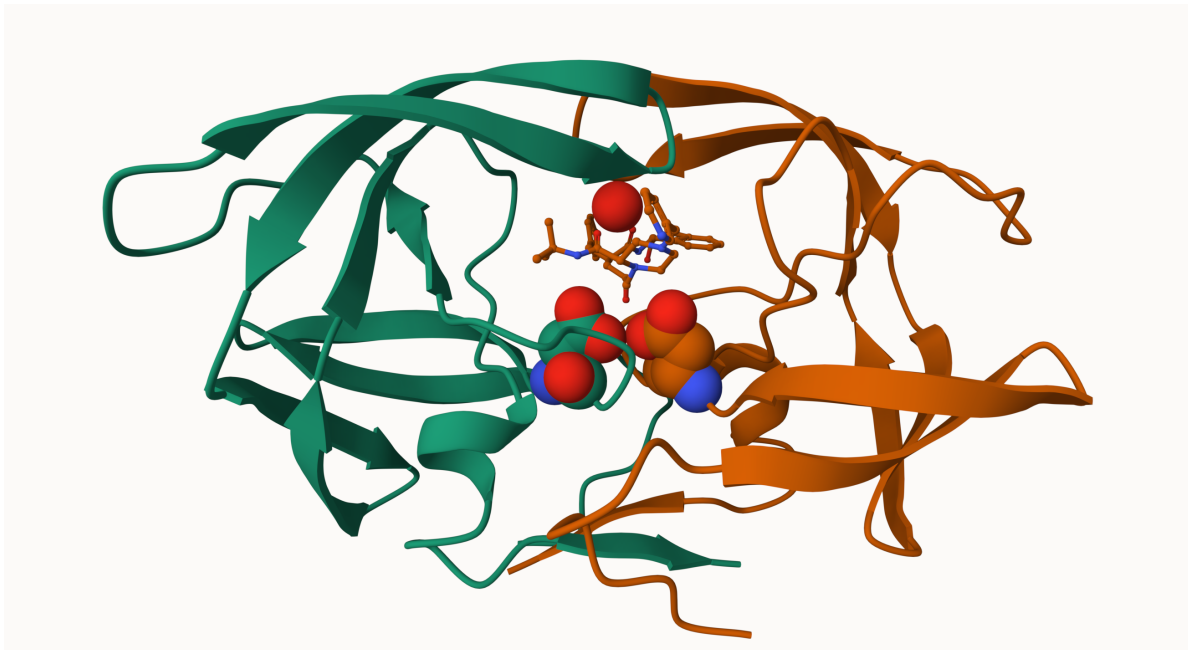


> Q4: Water molecules normally have 3 atoms. Why do we see just one atom per water molecule in this structure?

Q5: There is a critical “conserved” water molecule in the binding site. Can you identify this water molecule? What residue number does this water molecule have

Q6: Generate and save a figure clearly showing the two distinct chains of HIV-protease along with the ligand. You might also consider showing the catalytic residues ASP 25 in each chain and the critical water (we recommend “Ball & Stick” for these side-chains). Add this figure to your Quarto document.

A new clean image showing the catalytic ASP25 amino acids in both chains of the HIV-PR dimer along with the inhibitor and all important active site water.



Q7: [Optional] As you have hopefully observed HIV protease is a homodimer (i.e. it is composed of two identical chains). With the aid of the graphic display can you identify secondary structure elements that are likely to only form in the dimer rather than the monomer?

Bio3D Package for Structural Bioinformatics

```
library(bio3d)
```

```
pdb <- read.pdb("1hsg")
```

Note: Accessing on-line PDB file

```
pdb
```

```
Call: read.pdb(file = "1hsg")
```

```
Total Models#: 1
```

```
Total Atoms#: 1686, XYZs#: 5058 Chains#: 2 (values: A B)
```

```
Protein Atoms#: 1514 (residues/Calpha atoms#: 198)
```

```
Nucleic acid Atoms#: 0 (residues/phosphate atoms#: 0)
```

```
Non-protein/nucleic Atoms#: 172 (residues: 128)
```

```
Non-protein/nucleic resid values: [ HOH (127), MK1 (1) ]
```

```
Protein sequence:
```

```
PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGIGGFIKVRQYD  
QILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFPQITLWQRPLVTIKIGGQLKE  
ALLDTGADDTVLEEMSLPGRWKPKMIGGIGGFIKVRQYDQILIEICGHKAIGTVLVGPTP  
VNIIGRNLLTQIGCTLNF
```

```
+ attr: atom, xyz, seqres, helix, sheet,  
      calpha, remark, call
```

Q7: How many amino acid residues are there in this pdb object?

There are 198 amino acid residues in the pdb object

Q8: Name one of the two non-protein residues?

MK1 is one of the two non-protein residues

Q9: How many protein chains are in this structure?

There are 2 protein chains in this structure

```
head(pdb$atom)
```

	type	eleno	elety	alt	resid	chain	resno	insert	x	y	z	o	b
1	ATOM	1	N	<NA>	PRO	A	1	<NA>	29.361	39.686	5.862	1	38.10
2	ATOM	2	CA	<NA>	PRO	A	1	<NA>	30.307	38.663	5.319	1	40.62
3	ATOM	3	C	<NA>	PRO	A	1	<NA>	29.760	38.071	4.022	1	42.64
4	ATOM	4	O	<NA>	PRO	A	1	<NA>	28.600	38.302	3.676	1	43.40
5	ATOM	5	CB	<NA>	PRO	A	1	<NA>	30.508	37.541	6.342	1	37.87
6	ATOM	6	CG	<NA>	PRO	A	1	<NA>	29.296	37.591	7.162	1	38.40

	segid	elesy	charge
1	<NA>	N	<NA>
2	<NA>	C	<NA>
3	<NA>	C	<NA>
4	<NA>	O	<NA>
5	<NA>	C	<NA>
6	<NA>	C	<NA>

```
library(bio3dview)
library(NGLViewerR)
```

```
# view.pdb(pdb) |>
# setSpin()
```

```
# Select the important ASP 25 residue
sele <- atom.select(pdb, resno=25)

# and highlight them in spacefill representation
# view.pdb(pdb, cols=c("navy","teal"),
#           # highlight = sele,
#           # highlight.style = "spacefill") |>
# setRock()
```

Predicting Functional Motions of a Single Structure

```
adk <- read.pdb("6s36")
```

Note: Accessing on-line PDB file
PDB has ALT records, taking A only, rm.alt=TRUE

```
adk
```

```
Call: read.pdb(file = "6s36")
```

```
Total Models#: 1
```

```
Total Atoms#: 1898, XYZs#: 5694 Chains#: 1 (values: A)
```

```
Protein Atoms#: 1654 (residues/Calpha atoms#: 214)
```

```
Nucleic acid Atoms#: 0 (residues/phosphate atoms#: 0)
```

```
Non-protein/nucleic Atoms#: 244 (residues: 244)
```

```
Non-protein/nucleic resid values: [ CL (3), HOH (238), MG (2), NA (1) ]
```

```
Protein sequence:
```

```
MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMRLRAAVKSGSELGKQAKDIMDAGKLV  
DELVIALVKERIAQEDCRNGFLLDGFPRTIPQADAMKEAGINVDYVLEFDVPDELIVDKI  
VGRRVHAPSGRVYHVKNPPKVEGKDDVTGEELTTRKDDQEETVRKRLVEYHQMTAPLIG  
YYSKEAEAGNTKYAKVDGTPVAEVRADLEKILG
```

```
+ attr: atom, xyz, seqres, helix, sheet,  
      calpha, remark, call
```

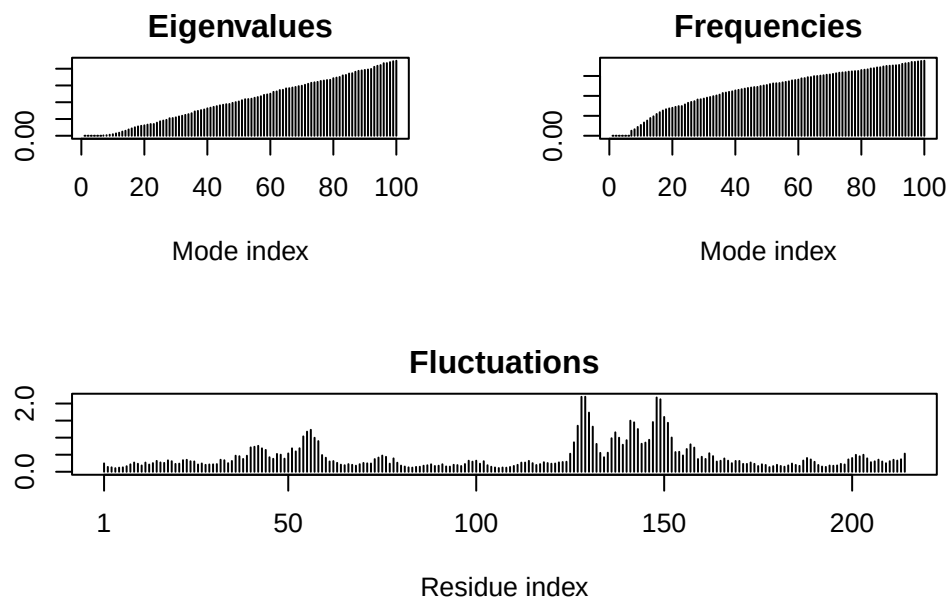
```
# Perform flexibility prediction
```

```
m <- nma(adk)
```

```
Building Hessian... Done in 0.01 seconds.
```

```
Diagonalizing Hessian... Done in 0.049 seconds.
```

```
plot(m)
```



Write out our results as a wee trajectory/movie of predicted motions:

```
mktrj(m, file="adk_m7.pdb")
```